AI-Powered Sign Language Learning & Assessment Platform

Workflow Design & Specification Document

Project Team:	Team 4 - Infosys Springboard Internship 2026
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Assigned Module:	Plan Application Workflows
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This design document outlines the step-by-step logic and visual workflows for how users interact with the sign language learning platform. It details onboarding, camera-based practice loops, scoring formulas, and error handling in simple terms.

1.1 Purpose of the Learning Workflows

The learning workflows describe the visual pathways and backend processes that make our platform function. This document helps the frontend and backend developers build a seamless user experience, guiding users from registration to daily sign practice and final certification.

1.2 Main Goals

- **Map out User Journeys:** Explain how a user signs up, starts a lesson, and gets evaluated.
- **Explain the Camera Practice Loop:** Detail how the browser reads webcam video, tracks hand shapes using MediaPipe, and shows instant feedback.
- **Define Database Storage:** Highlight when and where the database reads or writes data during user activities.
- **Draft Error Scenarios:** Create clear paths to recover when the camera is blocked, lighting is poor, or internet drops.
- **Incorporate Progression Logic:** Set rules for rewarding XP points, streaks, badges, and recommending next lessons.

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This table explains the 12 stages a learner goes through in the app, detailing user actions, system responses, and database writes in simple words.

#	Stage	What User Does	What System / AI Does	Database Write Action
1	Registration	Signs up or logs in	Checks login; issues JWT token	Creates new user row
2	Profile Setup	Enters goals & language	Saves user choices	Writes to learner_profiles
3	Diagnostic	Signs simple prompt words	Estimates baseline skill level	Writes to assessment_results
4	Path Select	Selects recommended path	Unlocks matching lessons	Writes to learner_path
5	Lesson Select	Clicks on a lesson card	Loads text and videos	Reads lessons, modules
6	Content View	Watches instruction video	Tracks playback progress	Writes to content_progress
7	Camera Practice	Makes the sign on camera	Runs hand tracker & AI check	Writes to practice_history
8	Live Feedback	Sees red/green outlines	Gives instant error correction	Writes to feedback_log
9	Graded Quiz	Takes an unguided test	Scores gesture accuracy	Writes to quiz_scores
10	Dashboard View	Checks overall score/stats	Compiles metrics for charts	Reads progress_tracking
11	Unlock Next	Proceeds to next level	Checks completion limits	Writes to unlocked_lessons
12	Certify	Passes final exam	Creates verified PDF download	Writes to certificates

3.1 New Learner Onboarding Flow

When a user signs up, they pick their role (like Learner) and set their daily study goals (like 15 mins a day) and experience level. The system saves this in the database to build their profile and recommend a starting point.

3.2 Returning Learner Resume Flow

Returning users skip setup. The app reads their saved login token, pulls their streak count and past progress from the database, and redirects them to the lesson they were working on last.

3.3 Practice Mode (Ungraded)

Practice Mode is ungraded and repeatable. The webcam runs in a loop to track the user's hand and show helpful colored outlines (green for correct, red for incorrect) alongside text tips. Practice attempts do not count toward quiz grades.

3.4 Assessment Mode (Graded)

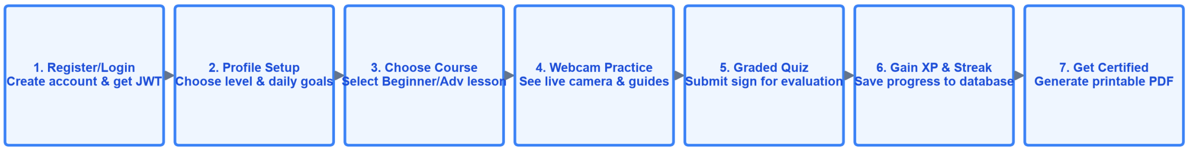
In Assessment Mode (or quizzes), all helpful green/red outlines and corrective tips are turned off. The system records the user's camera feed, classifies their sign, and writes the accuracy score directly to the database. Passing unlocks the next lesson.

3.5 Revision Mode (Custom Review)

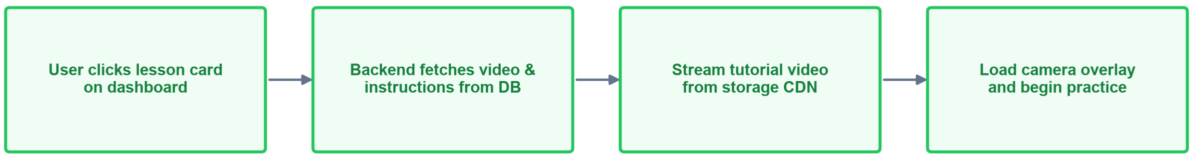
This mode automatically creates a review session for letters or words that the user struggled with in previous sessions (based on low scores logged in the feedback history).

Below are the workflow flowcharts showing the main user journey and lesson fetch pipeline.

Main Application Workflow Flowchart



Lesson Fetching & Demonstration Flowchart



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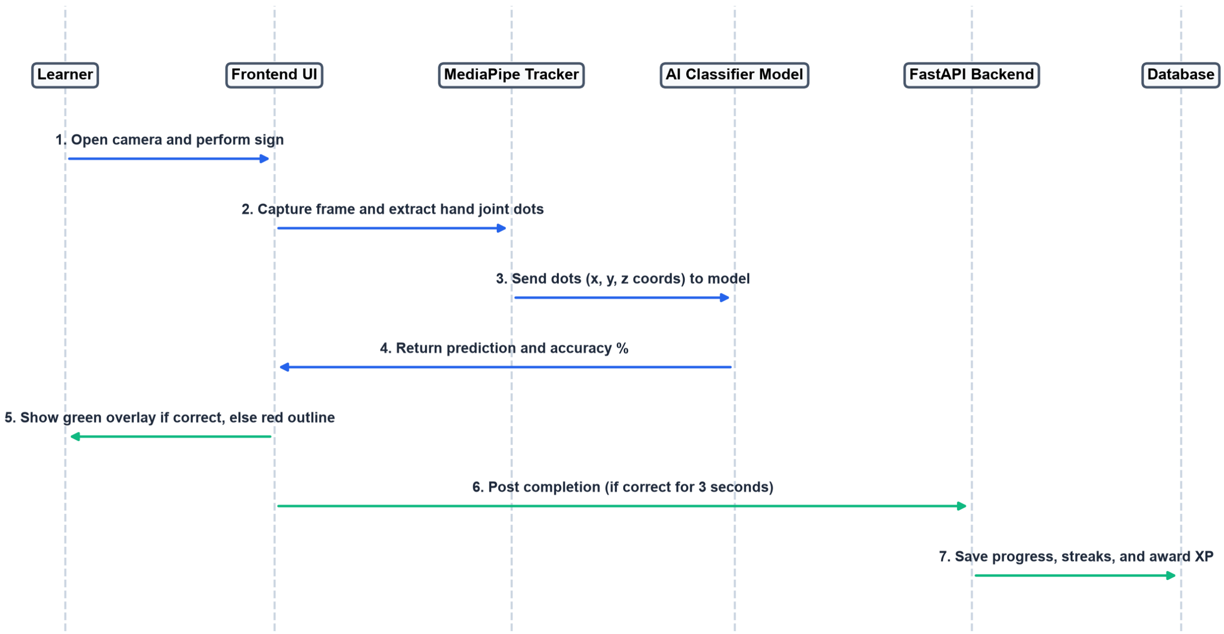
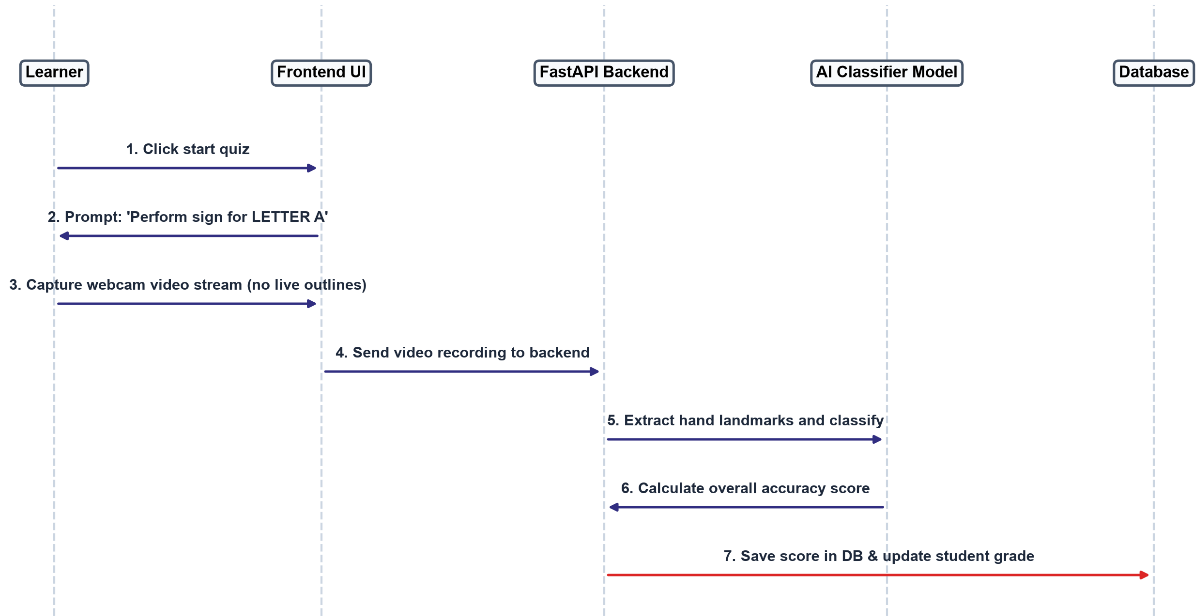
The practice loop captures frames from the webcam, tracks 21 points on the hand, classifies the gesture, and returns immediate green/red outlines.

Webcam Practice Loop Sequence Diagram

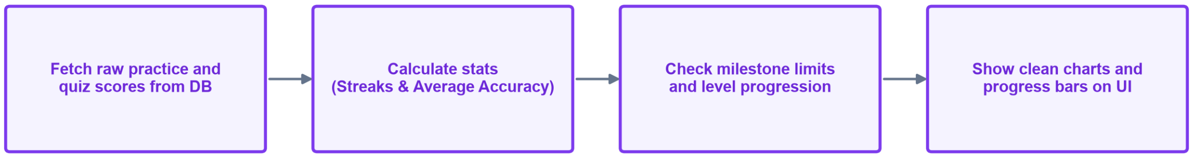


The quiz workflow evaluates student sign accuracy under testing conditions and compiles completion percentages to update dashboard metrics.

Graded Quiz & Video Assessment Sequence Diagram



Progress Tracking and Score Analytics Flow



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This section outlines what details are read from or written to database tables during the key workflow phases.

Workflow Stage	Database Table	Fields Saved or Retrieved
Registration / Login	Users Table	Saves user_id, email, password_hash, and role
Profile Setup	Learner_Profile Table	Saves level selection, goals, and preferred language
Diagnostic Assessment	Assessments Table	Saves diagnostic scores, completion timestamps, and level
Lesson Content Load	Courses, Lessons Tables	Retrieves lesson_name, course order, and tutorial content
Camera-Based Practice	Progress_Tracking Table	Saves completion_percentage and last_updated timestamp
Real-Time Feedback	Feedback Table	Saves feedback comments, rating, and date submitted
Quiz / Final Exam	Assessments Table	Saves final quiz score, date taken, and pass/fail status
XP / Daily Streaks	Learner_Profile Table	Updates streak_count, XP total, and list of unlocked lessons

Real-Time AI Pipeline Stages

- **Webcam Stream:** Browser captures video frame sequences at 25-30 frames per second.
- **Landmark Tracking:** MediaPipe Hands locates the hand shape and maps it to 21 joint keypoints (x, y, z coordinates).
- **AI Classification:** FastAPI loads the CNN model (for letters) or LSTM (for words) to compare user hand joints with reference joints.
- **Feedback Render:** System outputs a score. Green outline is drawn on camera screen if score is >= 80%, else a red outline with fix-tips.

Error Scenarios & Automated Recovery Actions

Failure Scenario	System Detection Point	Automated Recovery Action
Webcam blocked / denied	Browser MediaDevices check fails	Show warning alert; prompt user to grant permission in browser
Room is too dark	Hand tracking confidence falls < 40%	Overlay prompt suggesting better lighting or adjusting angles
Hand out of camera bounds	MediaPipe registers 0 tracked points	Draw hand placement box guide on screen; pause tracking
Internet connection drops	WebSocket connection fails / times out	Show offline banner; cache practice history in browser memory
Low confidence prediction	Classifier probability score < 60%	Withhold grading; prompt to try again or watch tutorial video

Gamification & Score Rules

- **Daily Streaks:** Increments by 1 every day the learner completes a practice. Resets to 0 if a full calendar day is missed.
- **XP Point Scoring:** Completing a lesson yields 10 XP, passing a quiz yields 30 XP, and final certification exam yields 100 XP.
- **Overall Score Formula:** Overall Score = 40% (Gesture Accuracy) + 25% (Quiz Marks) + 15% (Completion %) + 10% (Streaks) + 10% (Rate of Improvement).

Adaptive Recommendation Rules

- **Accuracy Thresholds:** If average accuracy falls below 60%, intermediate lessons are locked, and basic review exercises are suggested.
- **Milestone Skips:** If accuracy exceeds 90% with low response times, the system suggests skipping ahead to advanced blocks.
- **Mistake Review Lists:** Failed letters or words are logged and pushed to the top of the user's review dashboard.